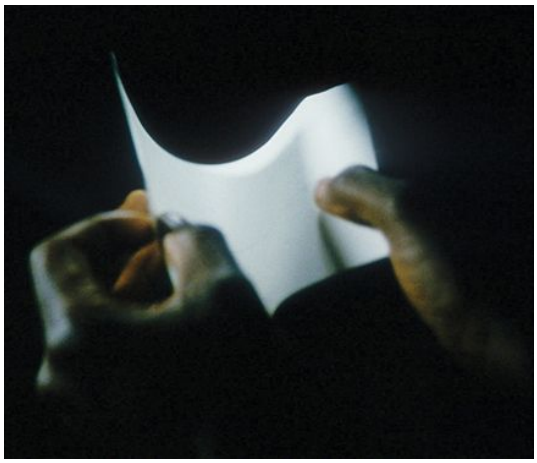
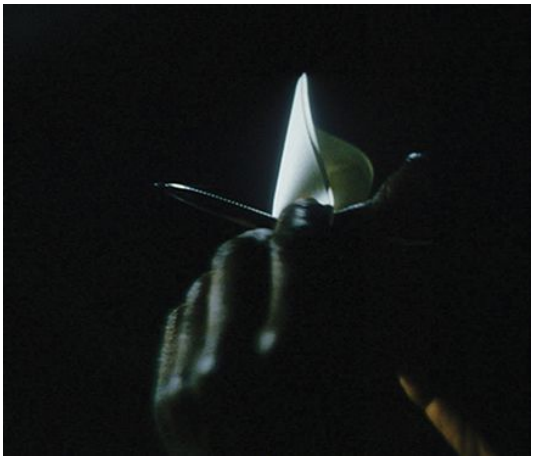



Fig. 14.9. Romilly explaining wormholes. *Left*: He bends a sheet of paper. *Right*: He punches a pencil (the wormhole) through the paper, joining its two edges. *[From* Interstellar, *used courtesy of Warner Bros. Entertainment Inc.]*

The Bottom Line

I doubt the laws of physics permit traversable wormholes, but this may be pure prejudice. I could be wrong. If they *can* exist, I doubt very much that they can form naturally in the astrophysical universe. My only real hope for forming them is artificially, in the hands of an ultra-advanced civilization. But we are extremely ignorant of how such a civilization could do it. And it appears more than daunting, at least from inside our brane (our universe), even for the most advanced of civilizations.

In *Interstellar*, however, the wormhole is thought to have been made, held open, and placed near Saturn by a civilization that lives in the bulk, a civilization whose beings have four space dimensions, like the bulk.

This is terra *extremely* incognita. Nevertheless, I discuss bulk beings in Chapter 22. In the meantime let’s talk about the wormhole in *Interstellar*.